

The use of high-frequency data in cryptocurrency research: A meta-review of literature with bibliometric analysis

Abstract

As the crypto-asset ecosystem matures, the use of high-frequency data becomes increasingly common in decentralized finance literature. Using bibliometric analysis, we characterize the existing cryptocurrency literature that employs high-frequency data. Based on 189 articles collected from the Scopus database from 2015 to 2022, we highlight the most influential authors, articles, and journals. This approach enables us to identify the emerging trends and research hotspots with the aid of co-citation and cartographic analyses and shows the knowledge expansion through authors' collaboration in the cryptocurrency research with co-authorship analysis. We identify four major streams of research: (i) return prediction and measurement of volatility in cryptocurrencies; (ii) (in)efficiency of cryptocurrencies; (iii) price dynamics and bubbles in cryptocurrencies; and (iv) diversification, safe haven and hedging properties of Bitcoin. We conclude that the investment features and the economic outcomes of highly trading cryptocurrencies would be analysed predominantly on tick-by-tick bases in the future. This paper provides recommendations for future research.

Keywords: Cryptocurrencies, high-frequency data, intra-day data, bibliometric analysis, network analysis, meta-literature review.

1. Introduction

With the advent of advanced high-frequency data analysis techniques, finance scholars have increasingly delved into the intricate dynamics of financial markets, thereby uncovering novel insights into their behavior and interconnectivity. However, for relatively new and immature financial assets, such as cryptocurrencies, the challenge of accessing high-frequency data posed a significant obstacle. As a result, a majority of the early papers in this field resorted to employing daily data, inadvertently missing out on the advantages of utilizing data with higher frequencies (Corbet et al., 2018a). Nevertheless, as the infrastructure supporting cryptocurrency trading matured, it became possible to collect data at higher frequencies, ushering in a new wave of cryptocurrency research that leveraged this more granular data (e.g., Katsiampa et al. 2022).

Contrary to low-frequency data, the high-frequency data helps in identifying jumps in financial assets (Christensen et al. 2014), facilitates the detection of behavioural biases, accurate information about market microstructures, intraday changes, and patterns that cannot be easily captured but are concealed at daily frequency (see Hansen & Huang 2016). The use of high-frequency (hourly) data help to search for arbitrage trades in both short and long-time horizons (Sifat et al. 2019). The inefficiency in cryptocurrencies can also be detected better in high-frequency (intraday) data than the low-frequency (daily) data (Zargar & Kumar, 2019). Moreover, low-frequency data, even daily data, cannot reflect the swiftly changing environment in crypto markets during trading. In contrast, high-frequency data that covers shorter time horizons can improve the precision of models and the understanding of the market dynamics i.e., monitoring the price dynamics and formation, trading activity, microstructures, and investor behaviours in the crypto market. From COVID-19 onwards, the literature evaluates cryptocurrencies at high frequencies from various perspectives (Katsiampat et al. 2022).

The results reported in cryptocurrency research papers, often developed in parallel, creating discrepancies that stem not only from differences in observation periods but also from variations in the data frequencies utilized by scholars. Alexander & Dakos (2020) finds that 80 out of 124 papers published between January 2017 and March 2019 use questionable and/or non-traded data in empirical analysis. To the best of our knowledge, no existing studies have previously attempted to comprehensively explore the cryptocurrency literature using high-frequency data, synthesizing the available findings and providing meaningful directions for future research in this domain. To

fill this gap and enhance the understanding of the role of high-frequency cryptocurrency data in shaping the cryptocurrency literature, this paper employs a bibliometric analysis.

The bibliometric technique is the top trending approach for analyzing growing literature, assessing the scholarly contributions of diverse stakeholder and helps creating thematic maps for scholarly applications. Along with it, we also perform a content analysis. We attempt to answer the following questions: What are the prevailing publication trends in the use of high-frequency data in crypto-related research? Which highly influential articles leverage high-frequency data in crypto-related studies? Does a surge in authors or journals' number of publications lead to a proportional increase in the total number of citations and getting more attention? Does publishing in highly influential journals in this field significantly contribute to a rise in the number of citations of authors' article? What are the primary research streams encompassing past and current themes in cryptocurrency literature using high-frequency data? *What are the prospective avenues of research within this field?*

This literature review aims to provide a comprehensive exploration of the diverse academic articles that have utilized high-frequency data to unravel the complexities of cryptocurrency trends, market volatility, and the interrelationships among different digital currencies. By synthesizing and critically analyzing the findings from these pivotal studies, this paper aims to provide a nuanced understanding of the implications of high-frequency data analysis in shaping our comprehension of the ever-evolving cryptocurrency landscape.

Our paper contributes to previous studies in several important ways. Firstly, we conduct the performance or citations analysis from the notion of articles' production growth, contributing authors, institutions, countries and key journals. We also examine the impact of rising number of publications authors and journals on their citations and attention. Further, we assess and identify the fields relevant to crypto market research and its associated link with other assets that dominate the use of the high-frequency datasets. This topic is underexplored in the literature; thus, this paper aims to fill this gap by classifying the body of knowledge, contributing to rapidly growing cryptocurrency literature, and offering specific directions for further research to cryptocurrency scholars. Our paper highlights the benefits of using high-frequency data in cryptocurrency research and we believe facilitate further research in these innovative areas.

Secondly, we conduct a meta-review of the literature through a bibliometric and content analysis, employing both qualitative and quantitative methods to synthesize the research on the cryptocurrency market, with a particular emphasis on the growing utilization of high-frequency datasets. While several alternative approaches are available for categorizing existing papers based on their primary themes and methodologies, such as the systematic literature review (Corbet et al., 2019), the use of bibliometric analysis is swiftly gaining prominence in the field of finance. Therefore, our contribution builds upon previous studies that have utilized alternative methodologies, including the systematic literature review (Ballis & Verousis, 2022; Corbet et al., 2019) or the survey method (Fang et al., 2022; Holub & Johnson, 2018), by integrating both meta-analysis and bibliometric review approaches in our research.

Finally, the novelty of our paper lies in its distinction from the recently published bibliometric analyses of cryptocurrencies. For example, several studies have exclusively conducted bibliometric analyses focusing on Bitcoin (Merediz-Solà & Bariviera, 2019; Ramona et al., 2019; Shen et al., 2020), while others have explored cryptocurrencies from various perspectives, including their growth (García-Corral et al., 2022), economic implications (Yue et al., 2021), financial asset role (Jalal et al., 2021), adoption trends (Sousa et al., 2022), and the evolution from a technological to an economic standpoint (Jiang et al., 2021). Our current systematic literature review and bibliometric analysis, however, specifically emphasize the use of high-frequency data in cryptocurrency market research. To the best of our knowledge, no available bibliometric reviews have centred on this specific subject yet.

We find that the highest number of publications are by *Ahmet Sensoy* (author) and are published in *Finance Research Letters* (journal). The frequently publishing authors and journals are also influential, whereas level of attention is not attached to frequency of publications. The articles published in influential journals (high number of citations per paper) attract more citations. The use of high-frequency data in cryptocurrencies research focuses on four major streams. The first group focuses on return prediction and measurement of volatility in cryptocurrencies, which is expected given lack of established pricing models and high volatility in the crypto markets. The second focuses on the (in)efficiency of cryptocurrencies. The market efficiency is an inherent feature of markets' microstructure. The third examines price dynamics and bubbles in

cryptocurrencies, and the fourth studies hedging, diversification and the safe haven properties of Bitcoin.

The rest part of the paper is ordered as follows. Section 2 discusses the background literature, Section 3 explains the data and methodology, Section 4 describes the bibliometric results, Section 5 identifies the research hotspots in the field, Section 6 offers future research directions, and, finally, Section 6 concludes.

2. Literature Background

Cryptocurrency, initially a niche product within small online communities, has evolved into a digital medium of exchange prized for its economic efficiency and rapid transaction speeds. Its journey can be traced back to the 1980s when David Chaum introduced a groundbreaking encryption method that is still in use today for web-based security and serves as the foundation for modern electronic funds transfer (Bunjaku et al. 2017). Starting in the 1990s, several digital currencies were developed; however, many of them, such as Wei Dye's b-money, failed to gain traction. A notable exception is PayPal, which has gradually expanded its influence over time. Notwithstanding past attempts at creating virtual currencies, like E-gold and Liberty Reserve in 1996 and 2006, respectively, the title of the first enduring and most famous blockchain-based cryptocurrency belongs to Bitcoin, introduced by Nakamoto (2008). Prior to Bitcoin's inception, the concept of cryptocurrency was virtually non-existent, primarily due to concerns surrounding centralization and the legitimacy of digital currencies (Bakar et al. in 2017).

The introduction of Bitcoin has ignited substantial excitement within financial markets, and this phenomenon poses an existential risk to certain conventional finance functions (Härdle et al. 2020). Subsequently, new cryptocurrencies, such as Ethereum, Ripple, Litecoin, and stable coins like Tether, have also been introduced, potentially serving as competitors or alternatives to Bitcoin. According to Statista, the cryptocurrency landscape has experienced exponential growth, expanding from 63 in 2013 to a staggering 4,501 options available today. Among them, only the top 20 account for 90% of the worldwide market share, achieving global dominance (Ji et al. 2019).

From a notable paper by Nakomoto (2008) that introduced Bitcoin, a plethora of papers have been published in economics, finance, computer science, management and other fields. Cryptocurrency

has become a revolutionary payment tool serving as an alternative to traditional financial markets due to their sage of innovative technology (blockchain) and cryptography¹, thereby ensuring a move towards the evolution of digital finance.

These newly developed digital currencies have advantages over conventional currencies, such as decentralized transactions, auditability, and anonymity, to name just a few, but they also contain unique risks, such as uncontrolled network operations (Johnson et al. 2014). The recent COVID-19 outbreak has spiked the interest of retail and institutional investors in the crypto market which can be accessed from home through Fintech trading sites such as Binance, Robinhood, and others (Long et al. 2022). This heightened accessibility of digital assets has led to enhanced intraday trading in the crypto market. Moreover, improved ecosystems such as Cosmos and Polkadot facilitate crypto investors to make cross-blockchain trades, which were not viable before.

The discussion of cryptocurrencies has become a hot topic that has attracted readers from industrial and academic communities (Tschorsch & Scheuermann 2016). Initially, the researchers focused on understanding the blockchain technology (Zheng et al. 2018; Zyskind & Nathan 2015) and the potential property of Bitcoin as a substitute for traditional currencies in the field of economics (Böhme et al. 2015; Yermack 2015). The current literature analysing the financial features of the crypto market shows enormous growth from the perspective of economics and finance (see Chu et al. 2021; Kayal & Rohilla 2021; Shi & Shi 2021), because this market with the billions of dollars capitalization serves as a complementary currency with diversification features resembling an investible asset class (Bouri et al. 2017a; Carrick 2016; Gandal et al. 2018). Moreover, with the maturity of the blockchain space, thousands of cryptocurrencies are being traded worldwide (Corbet et al. 2019). However, these digital assets demonstrate some standard features such as heavy tails, long memory, volatility clustering, and adverse leverage effects (Fung et al. 2022), and are speculative investment tools, which means they are prone to explosiveness or speculative bubbles (Bouri et al. 2019a; Shahzad et al. 2022; Cheah & Fry 2015) due to the realization of the potential for generating higher returns by investors, thereby attracting a new group of speculators.

¹ Cryptography word originated from military and different intelligence agencies those employ various codes to protect the leakage of information, for further details along with advantages and disadvantages of cryptocurrencies see, Bunjaku et al. (2017).

Regardless of the increased accessibility of high-frequency crypto data, most studies still employ low-frequency data in the crypto market and aligned research areas. This fact is highlighted by a survey conducted by Bariviera and Merediz-Solà (2021), which affirms that the authors of most studies contemplate daily frequency as a natural data frequency, ignoring other options. Among the most relevant studies, only a few authors employ high-frequency data and consider it more efficient. Besides the findings of this survey, numerous studies have explored different aspects of cryptocurrencies, predominantly Bitcoin by using data of daily frequency such as bubbles (Cheah & Fry 2015), hedging capabilities (Dyhrberg 2016b), liquidity and efficiency (Wei 2018), long memory interdependency (Cheah et al. 2018), structural breaks (Thies & Molnár, 2018), transaction activity (Koutmos, 2018), tail risk (Borri 2018) and its dependence (Nguyen et al. 2020), long memory effects (Phillip et al. 2019), and price clustering (Urquhart 2017), volatility spillover (Yi et al. 2018), and cyber-attacks, contagion and spillover within crypto market (Caporale et al. 2021).

Distinctively, the crypto market offers 24-hour trading throughout the year. In the existing state of the traditional financial market, investors may hold assets for a short period of a few minutes (Glantz & Kissell 2013), and the crypto market is no different. Many online exchanges offer tools and services that enable cryptocurrency traders to make trades in minutes or seconds, resulting in high profits (Brauneis et al. 2022; Chu et al. 2019). Moreover, compared to other financial instruments, cryptocurrencies demonstrate higher intraday volatility (movements of 3-5% are common), and a large portion of high-frequency trades are pursued through algorithmic trading bots, which signify the use of high-frequency data in analysing cryptocurrencies (Chan et al. 2022; Zhang et al. 2019b).

The significance of using of high-frequency data is also supported by several researchers. For instance, Mensi et al. (2020) suggest that a high-frequency data can locate or trace fitful market activities over time. Contrarily, daily prices also differ in their calculation and may create issues. Vidal-Tomás (2021) affirms that daily prices are either closing prices at the last closing hour of a day or weighted averages of hourly closing prices of a day (24 hours), which are statistically different. The last hour daily closing prices may fail to explain the daily behaviour of cryptocurrencies due to characteristic volatility, and those prices may change the next minute. Meanwhile, the weighted average daily prices may distort trader behaviour as they cause a

reduction in the volatility of cryptocurrencies via averaging the high-frequency (hourly) prices, leading to a persistent time series. Recently, Yarovaya and Zięba (2022) confirm that the return and volume of cryptocurrencies are not very sensitive to variations in sampling intervals of high-frequency data (i.e., 5-, 10-, 20-, 40-min, and hourly) but to variations between weekly, daily, and intraday data.

In line with the highlighted importance of high-frequency data, the recent research is more inclined to using of high-frequency data in exploring different aspects of cryptocurrencies, such as extreme connectedness between return and volume of cryptocurrencies at higher tail risk (Chan et al. 2022), long memory in return and volatility (Arouxet et al. 2022), rough volatility (Takaishi 2020) market complexity (Drożdż et al. 2020), price clustering (Li et al. 2020; Quiroga-Garcia et al. 2022) potential profitability by using a momentum strategy (Chu et al. 2020), volatility spillover among cryptocurrencies (Sensoy et al. 2021) and exchanges such as Binance, Coinbase, etc., (Alexander et al. 2022; Ji et al. 2021).

There have been previous attempts to aggregate and review cryptocurrency literature. Ballis and Verousis (2022) studied the impact of behavioral factors on cryptocurrencies, while Corbet et al. (2019) examined the breadth of cryptocurrency-related research. Fang et al. (2022) summarized existing research articles on cryptocurrency trading, covering topics such as trading platforms, signals, strategies, and risk management, and highlighting related outcomes. Holub and Johnson (2018) mapped research on Bitcoin and other cryptocurrencies across various disciplines, including technical fields, economics, law, public policy, finance, and accounting, among others. However, to the best of our knowledge, none of the existing studies have provided a comprehensive literature survey of recent cryptocurrency research employing high-frequency data. Therefore, in this paper, we aim to fill this gap through the use of meta-analysis and bibliometric review approaches.

3. Data and Methodology

This study employs the Scopus database to search for keywords and extract relevant articles as a literature retrieval procedure. For data extraction, we can use all platforms such as Scopus, Web of Science (Clarivate Analytics), EconLit, IDEAS/REPEC, and Google Scholar but it would necessitate a huge data-cleaning process to combine all datasets (Corbet et al. 2019). Compared to WoS, and EconLit, the Scopus has broader coverage of journals after 1996. While having the same

feature against REPEC, it does not include the working papers. Information on citations is available in Scopus, WoS and REPEC but not in EconLit (Ruane & Tol, 2007). However, relying on Scopus than other databases is based on its distinct features i.e., broader coverage with citation information and more than 15000 peer-reviewed subjects (Levine-Clark & Gil, 2008), notably 30% larger than the WoS database (Comerio & Strozzi 2019), and allows performing a more subtle analysis compared to Google Scholar which provides limited bibliometric information for conducting bibliometric analysis.

The keywords are "crypto-market" or "cryptocurrenc*" or "crypto-currenc*" or "digital currenc*" or "Bitcoin" or "BTC" or "Ethereum" or "ETH" and "high-frequency" or "high frequency" or "intra-day" or "intraday" or "hour*" or "tick?data" or "5?min" or "30?min". This search generates 344 documents written in English and published from January 2015 to July 2022. However, we limit our search to articles published or in the press by excluding book chapters, conference papers, editorial, erratum, data papers, and review papers, and get 240 articles. Finally, by further scrutinizing the articles for irrelevant terms such as high-frequency band, closing hours, peak hours, work-hours, intraday snapshots etc., we get the final list of 189 articles as a literature sample from 2015 to 2022. This bibliometric dataset is analyzed through a bibliometric technique. The year-wise publication trend is shown in Figure 1. A significant increase in publication using high-frequency data is noted after 2018.

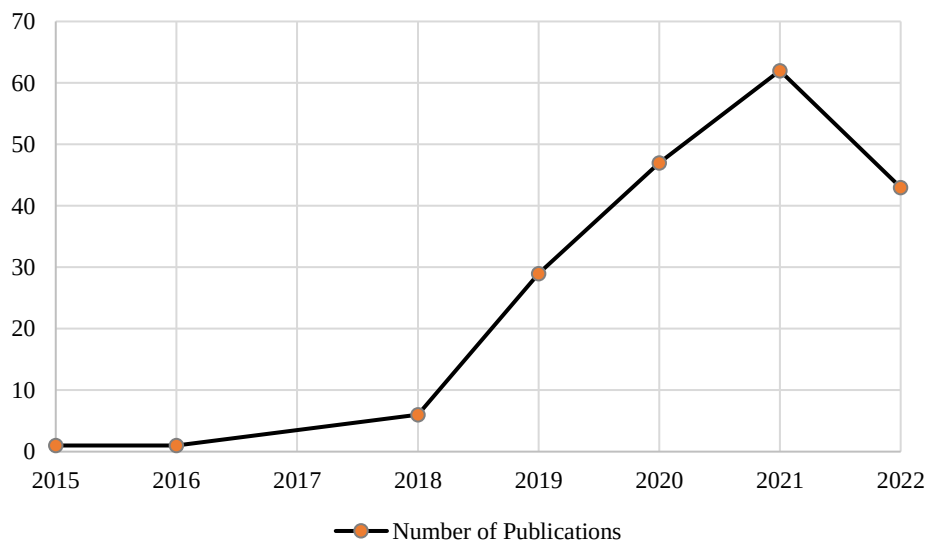


Figure 1. Publications trend over the years.

Bibliometric analysis has gained popularity in several fields and become a widely used approach to systematic review (Ruggeri et al. 2019; Patel et al. 2022). This approach can reveal hitherto unknown trends by collecting relevant information on a specific field or aspect of the subject. It can reduce subjective bias and steer authors to highly related fields to improve the quality of a literature review (Zupic & Čater 2015). Moreover, it helps researchers comprehend a wide picture and create links between publications from various sub-areas. Thus, it is suitable for identifying the existing state of research and future research avenues (Paul & Bhukya 2021).

Bibliometric research methods use quantitative statistical analysis on a set of publications, then maps in clusters of core authors, citations, and research institutions (De Bakker et al. 2005; Wang et al. 2021). It commonly focuses on citation, co-author, co-citation, cartographic analyses, and bibliographic coupling (Zupic & Čater 2015). This study uses bibliometric analysis and content analysis for a comprehensive examination of the use of high-frequency data in cryptocurrency research to offer insights into existing research interests and discover streams of research on emerging topics in the field. At the initial stage, the study employs citation analysis (no networks) to highlight the volume of publications and their influential role in a specific topic. This step includes a citation analysis for authors, documents, and journals. Further, a regression analysis is performed to determine the authors and journals' impact and the attention that they receive by publishing most frequently in this particular field. Also, the authors' impact while publishing in journals that are found influencing in this field. In the second stage, a co-authorship analysis is conducted to illustrate the network of collaborating authors in this field. In the third stage, a cartographic or keyword co-occurrence analysis is performed to identify the evolution of research streams on a particular topic. Furthermore, a co-citation (network) analysis is conducted to draw a scientific map that shows the general topics or streams of research through content analysis of each cluster.

Several software tools are frequently used to conduct bibliometric studies easier such as HistCite, CiteSpace, CiteSpaceV, Publish or Perish, SciMAT, BibExcel, VOSviewer, Biblioshiny, Pajek, and Gephi. Considering the diversity of features of different bibliometric tools, in this paper, we decide to use two software tools, namely, VOSviewer (van Eck & Waltman 2014) and Biblioshiny (Aria & Cuccurullo 2017). These fit to our aim of examining citation counts and classify literature on crypto market research utilizing high-frequency data, and offer the required analysis output.

These software tools are chosen due to their reliability in dealing with large datasets, comprehensive ability of sorting and dealing with manifold information types, flexible visual impairments, expedient settings, built-in algorithms and providing an eclectic range of other available network and visualization analytic tools such as Pajek and Gephi. Moreover, they are better than other potential software tools for different reasons. For example, CiteSpaceV can be an alternative, but it is not that much flexible and requires a lot of time to master its pretty complex user interface, while Histcite is confined to WoS and unable to interpret the Scopus dataset (Fahimnia et al. 2015), Bib Excel requires extra skills set and experience to conduct a modest analysis (Ampah et al. 2021), Pajek requires “.Net” dataset.

Therefore, software tools opted for this study are easy to use without requiring profound familiarity of clustering techniques and advanced computer skills and are freely accessible to anybody (particularly aspiring scholars) to identify existing trends and gaps in scholarly research articles, and publication databases. Biblioshiny contains a well-ordered and instinctive interface (Moral-Muñoz et al. 2020). We adopt it for descriptive and citation analysis including tables and graphs while we use VOSviewer for science mapping based on multiple reasons, such as data and network visualization, the clustering and the map created by it can easily be understood by readers from different disciplines. As identification of clusters in mapping and deriving themes out of them, VOSviewer is a compelling graphic interface for users that quickly observes the maps (Cobo et al. 2011). We also use Microsoft Excel for frequency analysis. We analyse the citation, co-citation, and keyword co-occurrence, thereby constructing a research atlas of the propensity for using high-frequency data in cryptocurrency and exploring the research flash points and frontlines in this field.

Tailored to aim of this study, we perform three key bibliometric techniques, i.e., co-author, co-citation, and keyword co-occurrence analyses, thereby constructing a research atlas of the propensity for using high-frequency data in cryptocurrency and exploring the research flash points and frontlines in this field. The roadmap of this study is shown in Figure 2.

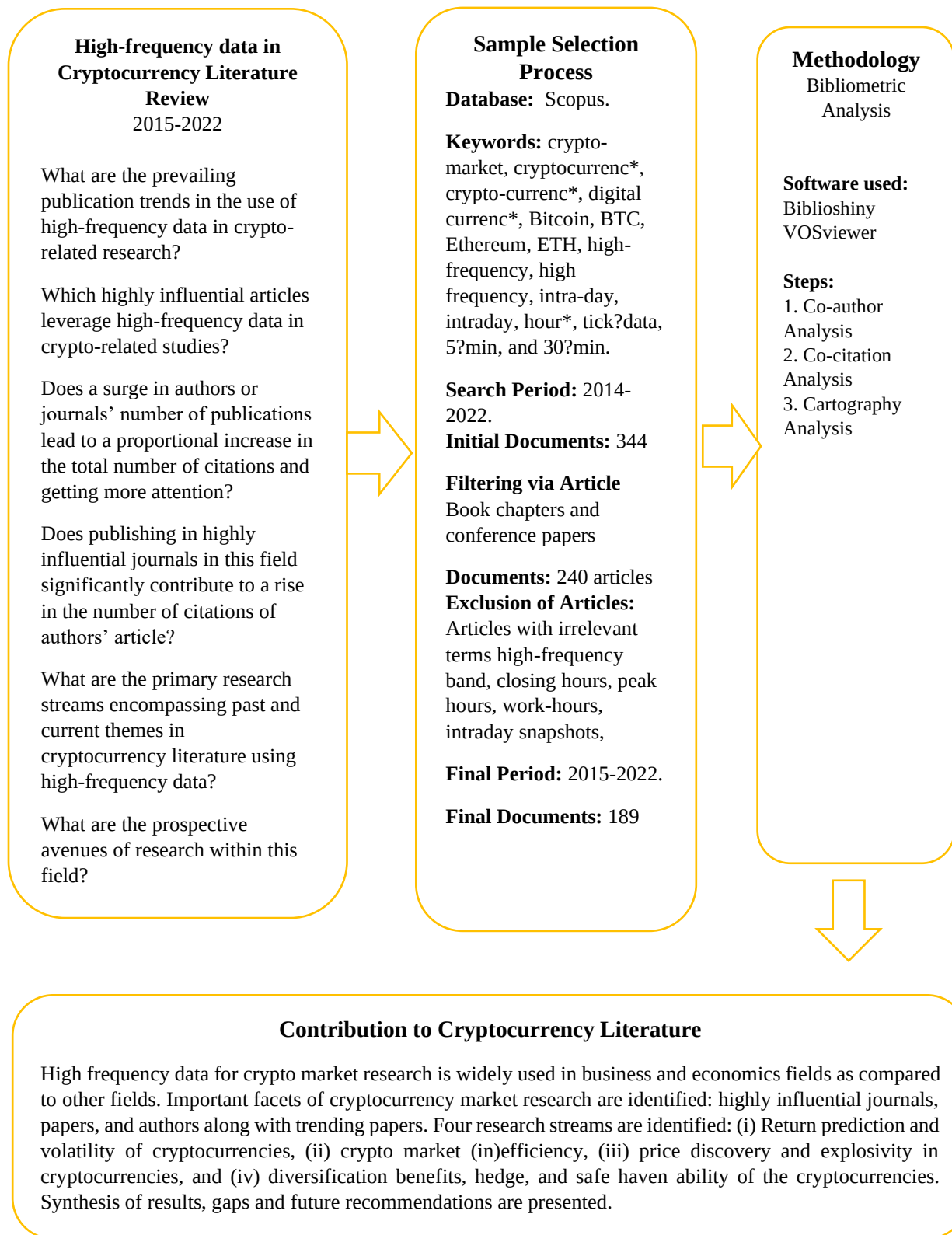


Figure 2. Road map from data extraction to analysis.

4. Bibliometric Results

This section reports bibliometric results: the performance analysis of authors, their affiliated institutions, nations, and the publishing journals in the form of citation tables using Biblioshiny, and network analysis including co-citation, co-authorship and keyword co-occurrence networks generated through VOSviewer.

We explain functioning of VOSviewer for network analyses. It is a compelling network visualization tool for structures comprising of several elements on the basis of distance-based envision approach. It creates networks or maps based on the data. Generated networks basically contain items or nodes representing objects of interest such as authors, sources, countries, keywords, etc., and lines that demonstrate the connection or relation between them (every pair of items) in the form of co-citation and co-occurrence, etc. (van Eck & Waltman, 2019). The strength of individual links is designated by a positive numerical value. The relation gets stronger corresponding to their higher value. The sum of the times the two items (e.g., authors) are quoted together indicates a link's strength. The items are grouped into clusters where every node is ascribed to exactly one cluster. While the number of resulting clusters depends on the clustering parameter such as cluster least size. This clustering analysis allows engraving of the sample into smaller and homogenous groups that can simply be recognized and explored. However, it was not the case to attain the highest homogeneity of separate clusters in our performed analysis. The clustering technique needs a relevant and specific algorithm to solve an optimization query or problem (for details see, Waltman, Van Eck & Noyons 2010). In doing so, VOSviewer utilizes smart local moving algorithm (Waltman & Van Eck, 2013). Therefore, the network analyses are performed through VOSviewer.

The figures or network diagrams generated through VOSviewer use clustering procedures to allot articles with higher strengths in creating results (network) and display them in different colors. As described earlier, the nodes or items represent authors, documents, keywords, etc., their size signifies highly cited or co-cited literature. Edges represent the relations in which the closeness of two nodes indicates the higher strength of the items (e.g., authors) and the co-citation or relation between them. However, they are distinguished from each other based on the variation in their total strength links or weighed-degree (computation is built-in feature of VOSviewer) or other centrality measures such as degree, betweenness and closeness centralities (can be computed through Gephi

and Pajek, etc.). Therefore, the network relationship between the items' (authors or documents, etc.) clusters helps determining the strength of connection among the clusters that create whole cluster analysis diagram of a discipline, field, or a specific topic (Li et al., 2017).

4.1. Performance Analysis

In this section, we perform performance analysis of authors, their affiliated institutions, and countries, and journals. We also categorize the top trending and highly influential articles published in this field. Furthermore, we examine the impact of authors and journals' number of publications on their citations and attention, and publishing in influential journals.

4.1.1. Citation Analysis

Citation analysis is a widely used method to assess the impact of a publication (Ding & Cronin 2011) through the citation numbers that an article receives from other authors' work (Zupic & Čater 2015). From 189 sample documents, 86 sources, and 395 authors, we report the top 11 authors, journals, and countries, their respective number of articles produced (Table 1), and their influence or higher number of citations (see Table 2) on the subject. We also categorize the highly cited (see Table 3) and top trending papers (see Table 4).

Table 1 shows a classification of the authors, journals and affiliated countries that are at the forefront of publishing high-frequency cryptocurrency market research. For instance, *Ahmet Sensoy* is the most prolific author in this area, with highest number of publications. The journal *Finance Research Letters* is the most frequent publisher of works on high-frequency cryptocurrency market research, with the *United Kingdom* being the top affiliated country.

Table 1. Highly publishing authors, journals, and affiliated countries.

Authors	TPA	Sources	TPS	AC	TPC
Ahmet Sensoy	8	Finance Research Letters	26	UK	30
Shaen Corbet	7	Physica A: Statistical Mechanics and Its Applications	13	US	27
Elie Bouri	7	Research In International Business and Finance	10	Germany	21
Paraskevi Katsiampa	5	International Review of Financial Analysis	9	China	20
Konstantinos Gkillas	5	European Journal of Finance	8	Canada	14
Andrew Urquhart	4	North American Journal of Economics and Finance	6	Turkey	13
Dirk Baur	4	Journal of International Financial Markets, Institutions and Money	5	Australia	11

Walid Mensi	4	Financial Innovation	4	Ireland	10
Stephen Chan	4	Applied Economics	4	Spain	10
Jeffrey Chu	4	Applied Economic Letters	4	Japan	10
Yuanyuan Zhang	4	Economic Letters	3	Greece	9

Note: TPA = Total publications of an author is a measure of the academic output of a researcher. TPS = Total number of publications of a source or journal. AC = Affiliated countries or a country from which the publishing author is affiliated. TPC = Total number of publications of authors affiliated with a country that is used as a metric to gauge the research and academic output of that country.

Table 2 presents a categorization of the authors, journals and countries of affiliation associated with highly cited articles on high-frequency cryptocurrency market research. For instance, *Ahmet Sensoy* is the most frequently cited author in the field. *Finance Research Letters* is the most influential journal for this topic, with the *United Kingdom* being the primary country of affiliation. From this list of papers, we can infer that there is a lot of research being done in the area of cryptocurrency pricing and trading, with a particular focus on Bitcoin. Researchers are exploring topics such as price discovery, forecasting volatility, price dynamics, correlations, inefficiencies, liquidity, transaction costs, and technical trading. This shows that the cryptocurrency market is an area that is gaining significant attention from researchers and investors, and that high-frequency data and trading techniques are being used to gain insight.

Table 2. Highly influential authors, journals, and countries.

Authors	TCA	Sources	TCS	AC	TCC
Ahmet Sensoy	297	Finance Research Letters	579	UK	615
Peter Molnar	249	Physica A: Statistical Mechanics and its Applications	398	Australia	412
Shaen Corbet	223	Journal of International Financial Markets, Institutions and Money	289	Turkey	322
Andrew Urquhart	193	International Review of Financial Analysis	265	Ireland	267
Paraskevi Katsiampa	135	Research in International Business and Finance	151	China	230
Dirk Baur	123	Economic Letters	103	Brazil	221
Walid Mensi	90	Chaos	99	US	197
Thomas Dimpfl	80	Journal of Futures Market	54	Germany	183
Stephen Chan	76	North American Journal of Economics and Finance	49	Spain	159
Jeffrey Chu	76	Financial Innovation	30	Viet Nam	157
Yuanyuan Zhang	76	European Journal of Finance	23	Canada	121

Note: TCA = Total citation of an author is a measure of the impact of a researcher. TCS = Total number of citations of a source or journal. AC = Affiliated countries or a country from which the publishing author is affiliated. TCC = Total number of citations of authors affiliated with a country that is used as a metric to gauge the research impact of that country.

Table 3 demonstrates the classification of highly cited or impactful papers that make use of high-frequency data, and cover a wide range of topics. For instance, Brandvold et al. (2015) explore price discovery on Bitcoin exchanges, which has had a major influence on cryptocurrency research, using high-frequency data. Urquhart and Zhang (2019) look at the hedge and safe haven abilities of Bitcoin for currencies using intraday data, which is the second most influential paper on cryptocurrency research that relies on high-frequency data.

Table 3. Highly influential papers.

Author	Title	Source	TC
Brandvold et al. (2015)	Price discovery on Bitcoin exchanges	Journal of International Financial Markets, Institutions and Money	158
Urquhart & Zhang (2019)	Is Bitcoin a hedge or safe haven for currencies? An intraday analysis	International Review of Financial Analysis	144
Peng et al. (2018)	The best of two worlds: Forecasting high frequency volatility for cryptocurrencies and traditional currencies with Support Vector Regression	Expert Systems with Applications	117
Chen et al. (2020)	Fear Sentiment, Uncertainty, and Bitcoin Price Dynamics: The Case of COVID-19	Emerging Markets Finance and Trade	108
Alvarez-Ramirez et al. (2018)	Long-range correlations and asymmetry in the Bitcoin market	Physica A: Statistical Mechanics and its Applications	108
Sensoy (2019)	The inefficiency of Bitcoin revisited: A high-frequency analysis with alternative currencies	Finance Research Letters	103
Aalborg et al. (2019)	What can explain the price, volatility and trading volume of Bitcoin?	Finance Research Letters	91
Katsiampa et al. (2019)	High frequency volatility co-movements in cryptocurrency markets	Journal of International Financial Markets, Institutions and Money	90
Dyhrberg et al. (2018)	How investible is Bitcoin? Analyzing the liquidity and transaction costs of Bitcoin markets	Economics Letters	73
Nakano et al. (2018)	Bitcoin technical trading with artificial neural network	Physica A: Statistical Mechanics and its Applications	59

Note: This list covers the top ten papers that are cited the most on the subject.

Table 4 classifies the top articles that employ high-frequency data, and encompass a variety of topics, based on their higher citations per year. Again, Urquhart and Zhang (2019) is the most trending paper on cryptocurrency research which explores the investment characters of Bitcoin as a hedge and safe haven. Chen et al.'s (2020) paper examining the impact of fear sentiment and uncertainty on Bitcoin price dynamics during COVID-19 is the second most widely referenced. In explaining the impact of articles, it is interesting to note that some articles reported in Table 3, e.g., Dyhrberg et al. (2018), and Nakano et al. (2018), are not reported in Table 4, since they have lower

citations per year, whereas two articles reported in Table 4, Baur and Hoang (2021) and Naeem et al. (2021), are not reported in Table 3, since they are not highly cited, but their per year citations are higher.

Table 4. Top trending papers.

Author	Source	Citations/year
Urquhart & Zhang (2019)	International Review of Financial Analysis	36.0
Chen et al. (2020)	Emerging Markets Finance and Trade	36.0
Sensoy (2019)	Finance Research Letters	25.8
Peng et al. (2018)	Expert Systems with Applications	23.4
Aalborg et al. (2019)	Finance Research Letters	22.8
Katsiampa et al. (2019)	Journal of International Financial Markets, Institutions and Money	22.5
Alvarez-Ramirez et al. (2018)	Physica A: Statistical Mechanics and its Applications	21.6
Brandvold et al. (2015)	Journal of International Financial Markets, Institutions and Money	19.8
Chen, Li & Sun (2020)	Journal of Computational and Applied Mathematics	18.3
Baur & Hoang (2021)	Finance Research Letters	18.0
Naeem et al. (2021)	Physica A: Statistical Mechanics and its Applications	16.0

4.1.2. Regression Analysis

We use the dataset consisting of authors and journals' number of publications, their total citations they received for articles and citations per article to run OLS regressions in which we examine the link of authors and journals publications count to their corresponding citations and attention as well as influential journal's role in getting more citations (see, Table 5).

In Table 5, we show the regression results in three different panels. In panel A, we show authors' increase in the number of articles significantly increases the number of citations, while reducing the citations per article. This implies that publishing more may improve the author's total citation counts but at the same time, it can adversely affect the average citations per article. The reduction in average citation per article may result due to multiple factors such as publishing on diverse research areas, variation in the importance of topics, e.g., groundbreaking one may get higher citations and reworking or that if not aligned with the prevailing research trends cannot receive the same level of attention. This result is generated using the total number of authors citations (right panel A of Table 5) and citations per article (left panel A of Table 5) as dependent variables and the number of publications as regressors. Similarly, in panel B, we repeat the same regressions with

journals' number of articles as the independent variables and journals' total citations (right panel B of Table 5) and average citations per journal (left panel B of Table 5) as dependent variables. We find that the rise in journals' number of publications enhances their citations, however, the results for attention or average citations per journal are insignificant. Finally, in panel C, we investigate the impact of publishing in highly influential journals (here we take highly cited journals publishing in this field) by running a regression for authors' total citations (independent variable) and journals' total citations. The results demonstrate that publishing in journals whose articles are most frequently cited due to publishing on similar topics or fields, or impactful content may result in an increase in authors' citations.

Table 5. Regressions for authors publication impact.

<i>Panel A: Publications 'impact on author's citations</i>					
Dependent variable: TCA			Dependent variable: C/A		
Variable	Citations		Variable	Attention	
	Estimated Coefficient	p-value		Estimated Coefficient	p-value
Constant	73.812	0.000	Constant	98.948	0.000
NP	12.131	0.002	NP	-14.769	0.000
<i>Panel B: Publications' impact on journal's citations</i>					
Dependent variable: TCJ			Dependent variable: C/J		
Variable	Citations		Variable	Attention	
	Estimated Coefficient	p-value		Estimated Coefficient	p-value
Constant	-12.930	0.084	Constant	12.686	0.000
NP	22.455	0.000	NP	0.569	0.376
<i>Panel C: Impact of influential Journal's publications on author's citations</i>					
Dependent variable: TCA					
Variable	Citations				
	Estimated Coefficient	p-value			
Constant	71.716	0.000			
TCJ	0.057	0.000			
Note: NP= Number of articles published, TCA =Total citations of authors, TCJ =Total citations of journals, C/A = Citations per article, and C/J = Citations per journal. Functions of Panel A are $TCA = f(TPA)$, and $C/A = f(TPA)$, for panel B, $TCJ = f(TPJ)$, and $C/J = f(TPJ)$. While function of panel C is $TCA = f(TCJ * \text{No. of articles published in journal})$.					

4.2. Network Analysis

4.2.1. Co-authorship Analysis

The co-authorship analysis enables us to create a link between authors, institutions and the countries that conduct research in a subject or topic (Van Eck & Waltman 2014). In our analysis of co-author networks using high-frequency data in crypto markets, we keep our criteria of a minimum of five documents per author that generates a network of fourteen authors distributing into four clusters (see Figure 3). The colour of the clusters varies, with cluster 1 in red, cluster 2 in green, cluster 3 in blue, and cluster 4 in yellow.

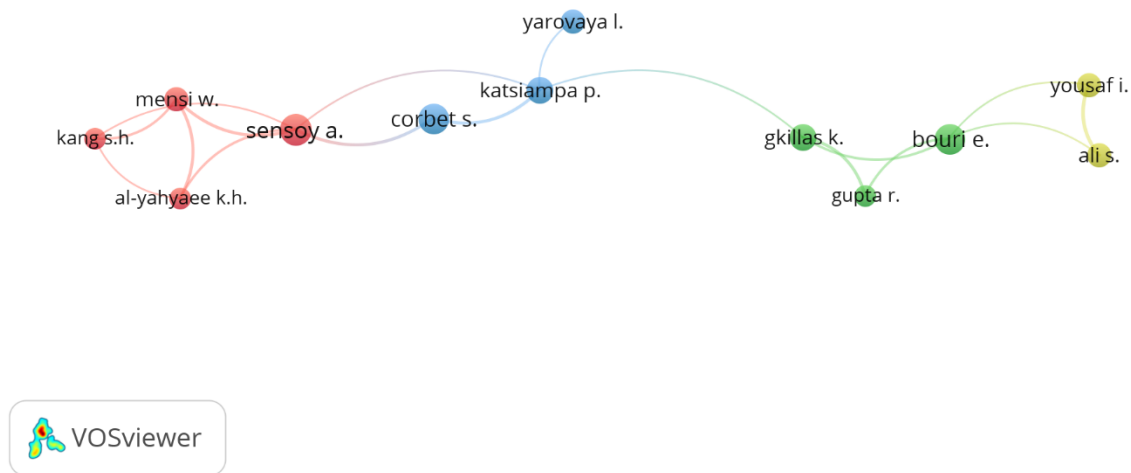


Figure 3. Co-author network of authors conducting research on cryptocurrencies.

4.2.2. Keywords Co-occurrence Analysis

Keywords are words given in a paper to reflect the content. Authors mostly index keywords based on the influence of their predecessors and the connections between the keywords in the article. When enough authors use similar keywords and accept those connections in their articles, it can be used to explore the topics or hot spots in a field (Whittaker 1989). Co-word analysis reveals the patterns or trends in any discipline by measuring the strength of linkages between the keywords, whereas the co-word network on the basis of the frequency of their co-occurrences reflects the content similarities or words with similar meanings (Ding et al. 2001). We create a keyword co-

occurrence network by selecting author's keywords with a minimum of five occurrences to identify the evolution of research flashpoints and embedded relevant topics.

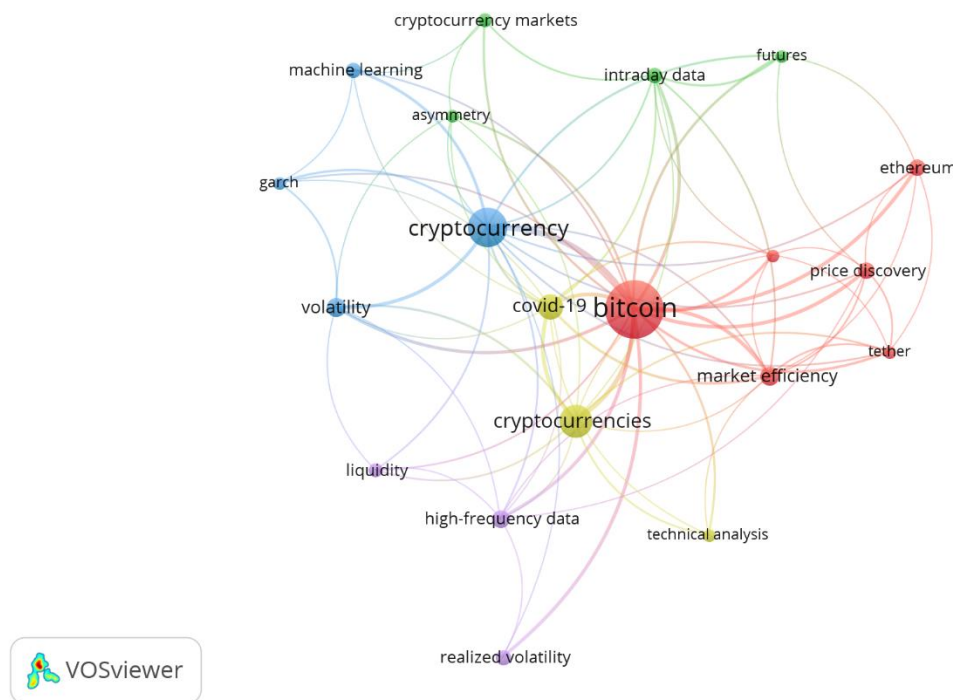


Figure 4. Keywords co-occurrence network of cryptocurrencies.

The keyword co-occurrence network generates five clusters (see Figure 4). The first cluster in violet has a major focus on assessing the volatility of cryptocurrencies, specifically Bitcoin, the second cluster in yellow focuses on price discovery and market efficiency of Ethereum and stablecoin i.e., Tether. The third cluster in red highlights the role of volatility and liquidity of cryptocurrencies using high-frequency data. The fourth cluster in green focuses on asymmetry in cryptocurrencies' futures. The fifth cluster in blue is work on the safe haven ability of Bitcoin and gold during COVID-19.

4.2.3. Co-citation Analysis

Co-citation analysis reveals the relationship between two papers if they are cited by any third paper concurrently (Small 1973). It shows the relation by distributing the documents with higher strength in the form of a network or clusters with the help of clustering methods. This relationship can examine the various research field commonality, centrality, and progression. We perform co-citation analysis using Vosviewer and select 20 as the minimum number of citations of an author.

This generates four clusters illustrated in different colours (see Figure 5). Cluster 1 in yellow highlights the focus of studies on return prediction and measurement of volatility in cryptocurrencies. Cluster 2 in green is studies focused on the informational (in)efficiency of cryptocurrencies. Cluster 3 in red is studies focused on the price discovery and speculative bubbles in cryptocurrencies. Cluster 4 in blue is research contributing to the field of portfolio diversification examining the co-movement, spillover, hedge and safe haven ability of cryptocurrencies against other assets.

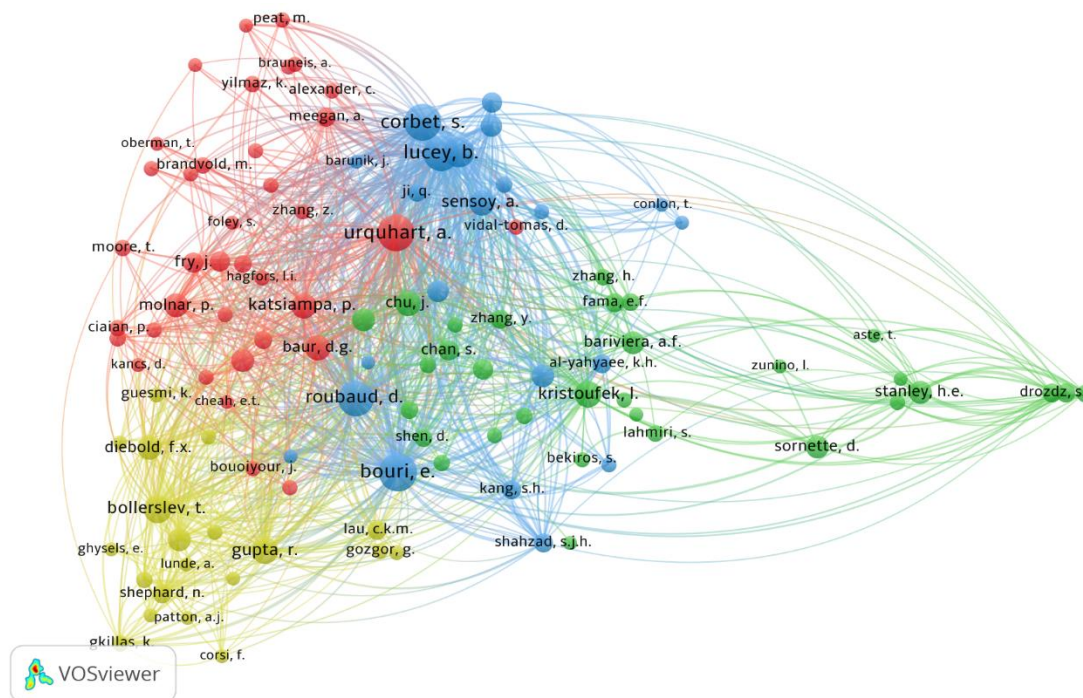


Figure 5. Co-citation network of authors conducting research on cryptocurrencies.

4.3. Identification of Major Streams of Cryptocurrencies Research

We perform content analysis (Potter & Levine-Donnerstein 1999) to reveal, arrange, and validate the research streams identified by bibliometric analysis. This analysis reveals the concepts, classifications, evolutions, and the current research streams of crypto market literature. Based on this, we identify four major cryptocurrency research streams through a co-citation network of cited authors (Figure 5). As noted by Persson, (1994), the citing publications constitute a research front while the cited publications form an intellectual base.

4.3.1. Return prediction and volatility in cryptocurrencies

This stream of research focuses on the peculiarities of return prediction and volatility forecasting in cryptocurrencies, with particular emphasis on high-frequency data and various volatility models.

Return prediction and volatility in cryptocurrencies have been studied since the introduction of Bitcoin by Nakamoto (2008). The volatile nature of cryptocurrencies is attributed to their newness compared to the equity markets (Katsiampa 2017). Various factors have been found to drive Bitcoin prices, such as searching sites (Kristoufek 2013), market forces (Ciaian et al. 2016a), or economic fundamentals (Li & Wang 2017), transaction speed (Corbet et al. 2018a), market attention (Urquhart 2018), trading volume (Balcilar et al. 2017; Bouri et al. 2019b), economic policy uncertainty (Demir et al. 2018; Fang et al. 2019), cryptocurrency liquidity (Wei 2018), Google search index (Bleher & Dimpfl 2019; Nasir et al. 2019), and Twitter tweets (Karalevicius et al. 2018; Shen et al. 2019). Besides prices, these factors have also been found to predict cryptocurrency returns, volume and volatility. Different models and, machine learning techniques, such as long-short term memory neural networks (Alonso-Monsalve et al. 2020; Lahmiri & Bekiros 2019; McNally et al. 2018) and support vector machines (Akyildirim et al. 2020), have been employed to predict cryptocurrency returns. Studies have shown that the liquidity of cryptocurrencies rely on their specific variables and are irrelevant to global financial markets variables (Brauneis et al. 2021). Studies have also analyzed crypto markets' liquidity by using high-frequency data (e.g., Dyhrberg et al. 2018; among others).

After the introduction of realized volatility based on high-frequency data by Andersen et al. (1998), it is used to study the stochastic properties of returns (Barndorff-Nielsen & Shephard 2002). In doing so, several volatility-based models i.e., ARCH (Engle 1982) and GARCH (Bollerslev 1986), the jump detection methods within the realized volatility by separating bi-power variation and realized variance (Andersen et al. 2007; Barndorff-Nielsen & Shephard 2003, 2004, Huang & Tauchen 2005), up and downside realized measures (semi-variances) entirely based on the up and downward intraday price fluctuations (Barndorff-Nielsen et al. 2006). The use of high-frequency data can accurately assess asymmetries in volatility over longer daily return horizons (Bollerslev et al. 2006). Later, methods filtering out realized rare and large jumps are also introduced (Tauchen & Zhou 2011). However, volatility or volatility jumps can be predicted by several factors such as geopolitical risk (Gkillas et al. 2018). Moreover, the very good and bad news draw its diverse

impact on volatility such as relatively good (intra-daily) news reduces the future or next day volatility (Chen & Ghysels 2011). The prediction of Bitcoin's realized volatility depends on the historical realized volatilities and jumps (Qiu et al. 2021). For instance, Patton and Sheppard (2015) argue that future volatility highly depends on the volatility of past negative returns and the sign of jump matters for the impact of price jump on volatility such as positive (negative) jumps lower (raise) future volatility. Moreover, the existence of co-jumping behavior in volatility of cryptocurrencies may spark future jumps (Gkillas et al. 2022).

Although, Corsi (2009) recommends the HAR-RV model as a better method that can reproduce the key empirical properties of financial returns (self-similarity, long memory, and fat tails). However, several studies (e.g., Dyhrberg, 2016a; Guesmi et al. 2019; Katsiampa, 2017; Yu, 2019; among others) examine the return prediction and volatility in cryptocurrencies using other volatility models (e.g., Bergsli et al. 2022; Chu et al. 2017; Diebold & Yilmaz, 2012, 2014; Hansen et al. 2012; among others). Studies also study volatility cascades in cryptocurrencies against US dollar (Gradojevic & Tsiakas 2021) and volatility spillover among cryptocurrencies (e.g., Yousaf & Ali 2020).

High-frequency data is an efficient way to predict cryptocurrency returns (Vidal-Tomás 2022). Correlation between volume and frequency is found to be inversely proportional (Yarovaya & Zięba 2022). Thus, future research topics include detecting price manipulation in cryptocurrencies at intraday level, using feature selection techniques to get improved accuracy, analysing the impact of hard and soft forks on intraday reversal and momentum, assessing time-based momentum strategy during price crashes and sharp recovery by using high-frequency data (Wen et al. 2022), and examining casual patterns between Bitcoin and other cryptocurrencies. Additionally, the volume-return relationship of cryptocurrencies (Balcilar et al. 2017; Yarovaya & Zięba 2022) can be examined with the inclusion of seasonality and or volatility effects.

Similarly, by using machine learning approaches such as recurrent neural networks and long-short-term memory, new research can predict Bitcoin's intraday price dynamics and identify its drivers (e.g., online sentiment, media hype). We also provide some prospective avenues offered by researchers of the field. For instance, Gkillas et al. (2020) suggest examining the macro-financial determinants of crypto market jumps, and Grobys and Huynh (2022) propose assessing the impact of jumps in stablecoins. While Gkillas et al. (2022) suggest assessing the impact of

cryptocurrencies and other financial assets' jumping behaviour (upside, downside, large and small jumps). Finally, forecasting volatility requires different models for different data frequencies. The in-sample results and out-sample forecasts can be reconciled at short-, medium-, and long-term horizons.

4.3.2. (In)efficiency in the cryptocurrency market

Studies have identified various factors and developed different models or approaches that make able to predict cryptocurrency prices which challenge the efficient market hypothesis (all information is already reflected in prices). Therefore, research on cryptocurrency price prediction has led to the investigation of the efficiency of the cryptocurrency market, which is based on the efficient market hypothesis. Even the increased liquidity also makes cryptocurrencies inefficient and less predictable (Brauneis & Mestel 2018).

Like other assets, stylized facts such as fat-tailed distributions, volatility clustering, long memory features, (Bariviera 2017; Bariviera et al. 2017; Chu et al. 2015; Cont 2001; Takaishi 2018; Urquhart 2017), adaptive patterns of long memory (Khuntia & Pattanayak 2020), interactions among returns, volatility, volume, and bid-ask spread (Eross et al. 2019), and asymmetric relations with price trends are observed in cryptocurrencies (Nadarajah & Chu 2017). There is also evidence of asymmetric reverting behavior in Bitcoin, strongly for negative returns, while excessive levels of persistence are revealed for positive returns (Corbet & Katsiampa 2020). In similar vein, studies examine and analyze the microstructure properties or their impact on cryptocurrencies (e.g., Scaillet et al. 2020), stylized facts including multifractal properties (Stavroyiannis et al. 2019) by utilizing high-frequency data and different methods such as volatility autocorrelation, multi scaling effects, etc., (Drożdż et al., 2018), including Detrended Fluctuation Analysis, and Hurst exponents (Zhang et al. 2019a), and herding behavior during the COVID-19 pandemic and other different situations (e.g., Choi et al. 2022; Jalal et al. 2020; Mandaci & Cagli 2022; Yarovaya et al. 2021). Moreover, Bitcoin may respond to negative news more than the positive one (Fang et al. 2019). Negative sentiments in the market can predict bitcoin returns, volatility, trading volumes and jumps. For instance, the Bitcoin returns, and volatility fall in response to the news relevant to crypto fraud and cyber-attacks (Rognone et al. 2020). Moreover, evidence indicates that the behavioral patterns in Bitcoin-related google searches and tweets amplify the herding whereas, EPU patterns and dependence between foreign exchange and equity markets dampen it (Philippas et al. 2020).

Thus, events like COVID-19 pandemic may steer the cryptocurrency markets to be away from the weak-form efficiency and make possible to predict future cryptocurrency prices based on the past information. Furthermore, the existence of long-range dependence in cryptocurrency price returns and its amplification during the COVID-19 period points the inefficiency (Naeem et al. 2022).

In efficient markets (Fama 1970), past information should be reflected in prices, however, Bitcoin's anti-persistence behavior (Takaishi 2021) indicates a frequent reversal in its moving direction, and asymmetric volatility induces inefficiency (Kristoufek 2018; Phillip et al. 2018), and thereby the possibility of achieving abnormal profits (Caporale et al. 2018). Even though, Bitcoin has predictive information for price trends of other cryptocurrencies (Wang & Ngene 2020). To study the efficiency of crypto markets, several approaches, such as the Hurst exponent (Carbone et al. 2004), multifractal analysis (Drożdż et al. 2010; Jiang et al. 2018), wavelet and detrended fluctuation analysis (Oświęcimka et al. 2006), detrending analysis (Oświęcimka et al. 2014), and multifractal detrended analysis (Kantelhardt et al. 2002) have been employed for various assets (Jiang et al. 2019; Wang & Wang 2021) and or cryptocurrencies (Alvarez-Ramirez et al. 2018; Aslan & Sensoy 2020; Bariviera 2017; da Silva Filho et al. 2018; Dimitrova et al. 2019; Kristoufek 2018; Takaishi & Adachi 2020; Takaishi 2018; Urquhart 2016; Zhang et al. 2018).

Along with the approaches used to study the multifractal properties of Bitcoin, the efficiency in the cryptocurrency market is examined through benchmark currency, varying sampling frequency, etc., (Bariviera 2017; Chu et al. 2019; Khuntia & Pattanayak 2018; Jiang et al. 2018; Sensoy 2019; Tiwari et al. 2018; Vidal-Tomás & Ibañez 2018). Despite the various perspectives, the primary conclusion of studies is that Bitcoin is inefficient (Kristoufek 2018; Mensi et al. 2019; Nadarajah & Chu 2017; Urquhart 2016) but may achieve weak-form efficiency over time (Chu et al. 2019; Gradojevic et al. 2022). Based on the evidence of rise in economic value and trading anomalies during COVID-19 pandemic, Wen et al. (2022) suggest that the traders can form a profitable strategy based on the historical intra-day returns, thereby inefficient cryptocurrency market.

As previous studies examined herding behavior in bitcoin future markets (e.g., Blasco et al. 2022), however, investigating stylized facts and empirical results of cryptocurrency futures and spot markets is important. In a similar vein, herding effects and their intensity should be studied across countries and markets, while considering impact of distinct. Examining macro-financial, sentiment, psychological, and uncertainty determinants of herding behaviour in the cryptocurrency market can

provide detailed insights. The efficiency of cryptocurrencies in different regimes (up- and downside trends and asymmetric behaviour) should be assessed, as well as comparison of the relative efficiency of other assets with cryptocurrencies. Variations in persistence and market efficiency should be evaluated during the outbreak of COVID-19 and Ukraine-Russia war, and anomalies and efficiency of the more- and less-liquid cryptocurrencies should be assessed. Furthermore, as suggested by Dong et al. (2022), the anomalous returns (displaying long or short legs) are different from those identified in the equity markets, and the fundamental economic mechanisms of the market anomalies can be explored in future research.

4.3.3. Price discovery and speculative bubbles in cryptocurrencies

In preceding section, we discussed that the EMH and asset pricing theory follow the assumption that the asset prices move around or mean revert to their fundamental value. While the presence of explosiveness or bubbles in asset prices also indicates significant departure from their fundamental value. Similarly, studies attempt to predict bitcoin price through different approaches such as network-based properties of blockchain or bitcoin transaction graph (e.g., Greaves & Au 2015). However, according to Tiwari et al. (2018) and Urquhart (2016), the Bitcoin's fundamental value is zero, and individual investors drive the market, in a way such that noise traders can distort the price discovery mechanism, leading to inefficiencies. So, along with return and volatility forecasting, the price explosiveness or bubbles in crypto currency market also indicate inefficiencies in the market. Moreover, the utilization of high-frequency data has proven to be a valuable tool for examining price formation, discovery and speculative bubbles.

This research stream indicates that the literature on cryptocurrencies is vast, with a range of studies exploring their higher volatility than conventional currencies (e.g., Katsiampa 2017; Dwyer 2015), and internally-driven nature (e.g., Baek & Elbeck 2015), making it highly speculative (Yermack 2015). Yet some argue Bitcoin is a currency or a synthetic commodity money with monetary value, not a speculative investment (Blau 2017, 2018; Selgin 2015), while it possesses both speculative and financial asset properties (Kristoufek 2015) and competes with other cryptocurrencies (Gandal & Halaburda 2016). Studying Bitcoin's price formation mechanism, Ciaian et al. (2016b) report speculators influence Bitcoin prices, but macroeconomic factors only affect them in the short term. Subsequently, Ciaian et al. (2018) argue that the Bitcoin's transaction and speculative demand both

significantly impact its price formation. While Hu et al. (2019) suggest that the prices are usually formed owing to the negotiations and strategic trading.

On the other hand, over time, the price deviation across exchanges varies (Brandvold et al. 2015), with price discovery driven by shocks (Makarov & Schoar 2019). Using data from several exchanges, Brandvold et al. (2015) is the initial study that examined Bitcoin's price discovery mechanism by applying the econometric approaches of Hasbrouck (1995) and Gonzalo and Granger (1995). They argue that the exchanges with the highest trading volume are the pronounced price leaders with respect to the information share and find the United States' BTC-e and Japan's Mtgox to be price setters or leaders. While Pagnotoni and Dimpfl (2019) find that price discovery largely takes place on China-based OKcoin, Baur and Dimpfl (2019) and Corbet et al. (2018b) find spot prices lead the process, whereas Fassas et al. (2020) and Kapar and Olmo (2019) find Bitcoin futures to be dominant. Giudici and Polinesi (2021) and Giudici and Pagnotoni (2020) find Bitstamp, Coinbase, and Bitfinex to be leading, and Kraken to be a follower exchange. Chang and Shi (2020) find Bitcoin to be a highly dominant cryptocurrency in driving price discovery at the market level.

Besides looking at the effects of speculators and macroeconomic factors in short term, researchers also examine the endogeneity of price dynamics, and the potential for bubbles. Such as the endogeneity of price dynamics is an important concept for understanding the dynamics of price bubbles, as it explains how the expectations of investors can play a role in driving prices. Endogeneity of price dynamics is the idea that investors' expectations of future prices are based on current prices, which can lead to a feedback loop of rising prices that result from increased expectation of future prices. The price behaviour in price discovery and bubbles in the Bitcoin market are two different sides of a debate. Like the explosiveness also distorts the price discovery process. Turanova (2017) holds that Bitcoin's price discovery mechanism is unlikely to operate without an extreme rise in its price, suggesting that its true value is higher than its current level. In contrast, Cheah and Fry (2015) argue that the intrinsic value of a cryptocurrency is zero and that its exponential growth is a bubble. As discussed earlier, the studies such as Bouoiyour and Selmi 2015, Bouoiyour et al. (2016), and Kristoufek (2013) also consider bitcoin as a speculative bubble than a currency.

Fry and Cheah (2016) find a negative bubble in Bitcoin and Ripple, with Ripple's spillover intensifying price drops in Bitcoin. Fry (2018) identifies bubbles in Bitcoin and Ethereum, and argues that boom-bust cycles could occur in the absence of bubbles. Cheung et al. (2015) spot multiple transitory and three major bubbles in Bitcoin, with their burst corresponding to major events in the market. Bariviera et al. (2018) apply permutation-information-theory quantifiers to study the high-frequency price dynamics of 12 cryptocurrencies during the bubble and bust in starting month of 2018. They find cryptocurrencies demonstrate distinct dynamics, indicating that they are not homogenous assets. Corbet et al. (2018a) report Bitcoin to be in a bubble phase after it crossed a \$1,000 value. Bouri et al. (2019a) examine the link between explosive periods in the cryptocurrency market and find multi-directional co-explosivity, not necessarily from older to newer or larger to smaller markets.

There is a space for future research to study the relative price discovery of stablecoins, gold-backed cryptocurrencies, and tokens. These assets can contribute to price discovery as well as to the crypto market efficiency. Moreover, the degree of price deviation and integration between these digital currencies during the onset of specific events such as COVID-19 and the Ukraine-Russia war may help broaden the understanding of overall crypto market behaviour. As proposed by Su et al. (2022), it is important to examine the dramatic variations in the price discovery of Bitcoin at an intraday level in the post-covid period. Explosivity or bubbles in the cryptocurrency markets (Cheung et al. 2015), co-explosivity (Bouri et al. 2019a), and price explosiveness and the role of tweets (Shahzad et al. 2022) have been examined. However, the bubble contagion or spillover between cryptocurrencies and other financial assets could be examined. The impact of supply and demand on the boom and bust of bubbles could be analysed, along with the use of supply and demand graphs to simulate boom and bust periods. The connection and coherency pre- and post-price explosiveness in the crypto market would be another avenue of future research, as is information dynamics tools involving microstructure in the cryptocurrency market, which may provide early warning signals of crashes. As suggested by Vasiliauskaite et al. (2022), price and liquidity information dynamics concerning microstructure in times of price explosiveness can be examined in the future.

4.3.4. Diversification, hedge and safe haven assets

In times of crises, economic and financial shocks distort return predictability and the price discovery process. The smaller cryptocurrencies may start following the larger ones (Vidal-Tomás et al. 2018), and herding behavior tend to increase with the uncertainty (Bouri et al. 2019c), increasing prices can also make for a potential bubble in the market that can burst anytime. Additionally, in the crypto currency market, the cryptocurrencies portray different patterns of the volatility spillover due to the difference in the structure of return and volatility clusters among them (Sensoy et al. 2021). The highly market cap cryptocurrencies such as Bitcoin, Ethereum, and Litecoin, act as both transmitters and receivers of volatility in the crypto market (Polat and Kabakçı Günay 2021), while low cap ones such as Madsafe Coin also may also act as its transmitters (Yi et al. 2018). Such transmission of shocks result in lower diversification benefits. In such differing situations, investors start adjusting their investments and opt for portfolio optimization or hedging strategies. Particularly, during periods of downturns, the rising demand for hedge and safe haven assets may drive up their prices, leading to inefficiency in the market. Moreover, for the cryptocurrency market, several facets such as the price formation process and limited liquidity may head for an unreliable safe haven asset (Smales 2019).

This research stream discusses the cryptocurrencies literature which includes the inclusion of cryptocurrencies in financial assets portfolio as diversifier, hedge and safe haven. In addition to the daily inter-dependencies within the cryptocurrency markets (e.g., Ciaian & Rajcaniova 2018), studies examine diversification opportunities by using high-frequency data (e.g., Katsiampa et al. 2022; Sifat et al. 2019; Yarovaya & Zięba 2022). Bitcoin investment possesses the most distinctive features, consisting of remarkably high average return and volatility and lower correlation with other financial assets. Its inclusion, even in a small fraction, dramatically improves the risk-return trade-off of a portfolio (Brière et al. 2015) and diversifies the portfolios (Guesmi et al. 2019; Platanakis & Urquhart 2020). As a financial asset, it is a useful instrument for portfolio and risk management, and market sentiment analysis based on its value storing and exchanging benefits (Dyhrberg 2016a). Cryptocurrencies offer diversification benefits against other financial assets at short-term investment horizons (Corbet et al. 2018c). In a similar vein, Guesmi et al. (2019) report that the inclusion of Bitcoin, gold, emerging equity markets, and oil in a portfolio can significantly decrease the portfolio risk. Bitcoin can offer hedging income.

Bitcoin is found to display hedging abilities similar to gold against market risk (Dyhrberg 2016b). It can act as diversifier and poor hedge for US dollar, stock, bond, gold, oil, and commodity markets, whereas its safe haven properties are especially visible in extreme down movements in Asian stock markets. The data frequency matters to Bitcoin investors, as its properties vary across time horizons (Bouri et al. 2017a). It acts as a strong hedge and safe haven against fluctuations in commodity market indices (Bouri et al. 2017b), however, Klein et al. (2018) find no hedging ability against a decline in equity prices, and it can hedge against uncertainties (Bouri et al. 2017c). In intraday trading, Urquhart & Zhang (2019) find that it can diversify the risk of the Japanese yen, Australian and Canadian dollars, hedge against the euro, Swiss franc, and pound, and act as a safe haven for the Swiss franc, Canadian dollar, and pound. Shahzad et al. (2019) find a weak safe haven ability of Bitcoin, commodities, and gold against global stock markets, with varying properties across equity market indices. In another study, Shahzad et al. (2020) find the distinctive safe haven and hedge properties of Bitcoin and gold for G7 stock markets. Bouri et al. (2020a) find evidence that Bitcoin, Ripple and Stellar serve as safe havens for US equity indices, whereas Monero and Litecoin do so for the US equity market index. Bitcoin has the least dependence on other assets, and a superior safe haven ability, compared to gold and commodities (Bouri et al. 2020b).

Recently, investors' concern over the COVID-19 outbreak has led to research into the role of Bitcoin as a safe haven asset. Corbet et al. (2020a) find that investors are more inclined to view digital currency as a safe haven, but a few studies (e.g., Conlon & McGee 2020; Corbet et al. 2020b) suggest cryptocurrencies may even increase contagion. Conlon et al. (2020) indicate that Bitcoin and Ethereum do not act as safe havens for most equity markets, though Tether does. Yousaf & Ali (2020) reveal that portfolios of Bitcoin, Litecoin, and Ethereum are costlier and more effective for hedging during the pandemic. Finally, Das et al. (2020) conclude that Bitcoin is better at hedging crude oil uncertainty than commodities and gold, but not the US dollar.

It is possible to examine the liquidity spillover in cryptocurrencies to identify the diversification benefits. This behaviour can also be examined across assets. The diversification opportunities can be identified by examining the linkages between cryptocurrencies and meme stocks, rare earth stocks, etc., considering various investment horizons. Since COVID-19 and its induced financial market turbulence have been the initial test of the safe haven ability of Bitcoin (Kristoufek 2020),

it is important to assess the safe haven property of Bitcoin by considering differential impacts of intensity and type of crises. Another possible line of future research could be an assessment of the hedging effectiveness and the safe haven ability of Bitcoin and other cryptocurrencies in the recovery phase of the COVID-19 pandemic and during the Russia-Ukraine war. The impact of health and war induced crises may provide new insights depending on the nature of events. Gkillas et al. (2020) suggest the inclusion of Bitcoin, gold, and oil in a single portfolio to analyse the diversification benefits while studying the spillover effects of skewness, kurtosis, and jumps. As Baur and Hoang (2021) find six largest stablecoins act as safe haven against for bitcoin, and Baumöhl and Vyrost (2020) find few stable coins acting as safe haven against other top cryptocurrencies. There are a lot of prospective avenues to examine the safe haven ability of stable coins and tokens (compared to nonstable cryptocurrencies) against other assets. The symmetric or asymmetric multifractality and detrended correlation analysis of safe haven assets including cryptocurrencies, stablecoins and other financial assets can be a potential topic to explore. Since cryptocurrencies are becoming less influential, except Bitcoin, while dApps and Protocols are getting much attention from investors (Katsiampa et al. 2022), upcoming research can focus on these distinctive cryptocurrencies in their market analysis.

5. Avenues of Future Research

The research opportunities opened by cryptocurrencies and Bitcoin in scientific domains have prompted an in-depth examination of their future path and influence. Various types of digital currencies, such as stablecoins, altcoins and gold-backed cryptocurrencies, combined with the gradual progress of legalization in some countries, have made researchers focus on exploring the financial properties of these currencies.

In this section, we offer a few future research questions that do not unconditionally fit into one of the four previously mentioned research streams: What are the causes of crypto currency market in 2022? How investors perceive the future standing of the crypto market? Is there a need to empirically revisit the pre-crashes research findings? What are the impacts of Bitcoin's hard and soft forks on intraday reversal and momentum? What are the causal patterns among Bitcoin and other cryptocurrencies? What is the impact of changing regulations on the cryptocurrency markets? What is the impact of the rise of decentralized finance (DeFi) on the cryptocurrency markets? What is the impact of institutional investors on the cryptocurrency markets? What are the effects of

cryptocurrency derivatives on the cryptocurrency markets? What is the impact of news events on the cryptocurrency markets? What are the effects of cryptocurrency mining on the cryptocurrency markets? What are the effects of cryptocurrency mining on the environment? What are the pricing models used by cryptocurrency exchanges? What is the role of social media in influencing the cryptocurrency markets? What are the effects of algorithmic trading on the cryptocurrency markets? What are the effects of “whales” on the cryptocurrency markets? What are the challenges that stakeholders of crypto market face and how can they overcome them?

6. Conclusions

In recent years, the intensified use of high-frequency data has revolutionized the analysis of cryptocurrencies, which are now actively traded 24/7 across global markets. We adopted a comprehensive bibliometric technique, leveraging the co-citation and keyword co-occurrence analysis of 189 pertinent papers extracted from the multidisciplinary domains in the Scopus database spanning from 2015 to 2022. Our findings have not only illuminated the expanding landscape of literature within the cryptocurrency market but have also shed light on the leading articles, eminent authors, prominent institutions, reputable journals, and influential countries actively contributing to this domain.

Through a meticulous analysis of the bibliometric network and a thorough content assessment of the scholarly articles, we have identified and delineated four primary research streams that shape the discourse within the cryptocurrency market. These include studies focused on (1) examining the intricate dynamics of return and volatility behavior, encompassing predictive models and the phenomena of abrupt fluctuations within cryptocurrencies, (2) analyzing the inherent inefficiencies within the cryptocurrency landscape, (3) investigating the mechanisms of price discovery, including its formation, potential manipulation, and the emergence of speculative bubbles within cryptocurrency markets, and (4) evaluating the diversification benefits, hedging strategies, and safe haven properties, particularly pertaining to the flagship cryptocurrency, Bitcoin.

In the course of our analysis, we have meticulously pinpointed critical gaps and unexplored avenues within each research stream, thereby providing scholars with comprehensive insights into potential future directions for research utilizing high-frequency data in the cryptocurrency domain. It is worth noting that, as highlighted by Zha et al. (2020), the study of investor behavior in the context of cryptocurrencies necessitates a multidimensional and integrated interdisciplinary

approach. Therefore, it is evident that the trajectory of cryptocurrency research will transcend the confines of any single discipline, necessitating collaborations across a diverse range of academic fields and methodologies.

It is imperative to acknowledge that this article, while offering significant insights into the research conducted using high-frequency data in the cryptocurrency market, is constrained by its reliance on the Scopus database and its focus on the 24/7 trading nature of cryptocurrencies. Moving forward, the scope of this study could be expanded to encompass a broader spectrum of financial assets and alternate databases, such as Web of Science, SpringerLink, and others, to capture a more holistic view of the evolving landscape within the realm of digital financial assets. As bibliometric studies continue to serve as influential tools for spotlighting the growing impact of scholarly works and journals within specific domains, our research lays a sturdy foundation for further explorations and advancements in the realm of cryptocurrency research and analysis.

The insights gleaned from our comprehensive analysis of high-frequency data in the cryptocurrency market carry significant practical and policy implications for a range of stakeholders. As the cryptocurrency landscape continues to expand and evolve, investors, financial regulators, and market participants can leverage the identified research streams and trends to make informed decisions, manage risks, and develop robust strategies that capitalize on the unique properties of digital assets. Furthermore, policymakers can utilize our findings to establish a more comprehensive regulatory framework that addresses the intricacies of the cryptocurrency market, thereby fostering transparency, stability, and investor protection. By embracing the emerging research trends and addressing the highlighted gaps, stakeholders can foster a more resilient and secure cryptocurrency ecosystem that aligns with the evolving demands of global financial markets. Additionally, our study underscores the significance of interdisciplinary collaboration and the integration of diverse analytical tools in the study of cryptocurrency dynamics, underscoring the need for a cohesive and interdisciplinary approach that incorporates insights from fields such as economics, finance, computer science, and data analytics to ensure a holistic understanding of the multifaceted nature of the cryptocurrency market.

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